PRATHAM JAISWAL

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19 merged PRs · 10 upstream repositories

SUMMARY

Systems and automation engineer. Nineteen merged pull requests upstream into `rust-lang/rustc_codegen_cranelift`, `google-deepmind/open_spiel`, `pytorch/ao`, `CCExtractor` and `CircuitVerse`, with further work open against `rust-clippy`, `wagtail` and `gatsby`. Currently builds ticket-automation and vulnerability-reporting systems for a distributed healthcare research organisation. Rust, TypeScript, Python, PostgreSQL, Docker.

OPEN SOURCE

rust-lang/rustc_codegen_cranelift — compiler backend

- **#1625 (merged)** Ported an optimised disk-cleanup workflow that resolved CI runner disk exhaustion and restored abi-cafe test reliability.
- **#1711 (open)** Fixed an ICE on `const` asm operands pointing at anonymous allocations (`const &0`): materialised the allocation via `constant::data_id_for_alloc_id` and threaded the module into `GlobalAsmContext`, matching the inline-asm path and converting the remaining `unreachable!()` into a `span_bug!`.
- **#1710 (open)** Added `vcvttps2dq` instruction support.

CCExtractor/ccextractor — subtitle extraction, C

- **#2378 (merged)** Root-caused wrong tx3g subtitle timings on the FFmpeg MP4 path. `ccx_mp4_process_tx3g_packet` and `ccx_mp4_flush_tx3g` derived times from the shared `fts_now` pointer, which interleaved tracks had already advanced and which sat at zero for the first subtitle. Rewrote both paths to use packet PTS plus sample duration.

google-deepmind/open_spiel — game-theory research library

- **#1455 (merged)** Replaced legacy TensorFlow 1.x deep-RL examples (`eva.py`, `exploitability_descent.py`) with modern PyTorch equivalents.

pytorch/ao — PyTorch quantization

- **#3709 (merged)** Documented `FqnToConfig` and mixture-of-experts parameter-quantisation workflows in the API reference.

CircuitVerse/cv-frontend-vue — circuit simulator, Vue/TypeScript

- **Six merged**: removed jQuery from `engine.js`, `setup.js` and `minimap.js`; converted `subcircuit.js`, `events.js`, `sequential.js` and the LSB/MSB modules to TypeScript during the v1 simulator migration. **#1448 (open)** continues the same conversion.

gatsbyjs/gatsby — **#39703 (open)**: `gatsby-plugin-manifest` now respects `assetPrefix` in icon paths.

aboutcode-org/scancode-toolkit — **#4765, #4766 (open)**: OSGi / `.bnd` dependency parsing; authenticated Git dependencies no longer mis-parsed as npm aliases.

Also merged: Razorpay payment flow and video quality scoring in private client repositories; telemetry, retry and persistent queue storage (scheduler); Google OAuth and JWT decoding (lost-and-found); unique-username and signup fixes (profileFolio).

EXPERIENCE

Braid-Forbes Health Research — Full-Stack Developer, IT Systems & Operations *Remote · Apr 2026 – Present*

- Built a ticket automation layer over monday.com and Outlook: intake through received / on-hold / completion transitions with state-driven notifications, including debugging of sender connections and template state.
- Built a vulnerability reporting pipeline that consolidates endpoint findings by application, severity and remediation path, turning raw ThreatDown/Malwarebytes Nebula output into a document staff act on.
- Administer Microsoft 365 for a distributed research staff — Exchange, Teams, shared mailboxes, distribution lists, licensing, onboarding and offboarding.
- Keep remote staff working across FortiClient VPN, FortiToken MFA, RDP, Windows VMs, NAS and TeamViewer; document repeatable fixes so the same ticket does not come back twice.

PROJECTS

Perfume Store — production commerce system · TypeScript, Next.js

Catalogue, admin console, dispatch and tracking, India Post integration, variant pricing, transactional email. Shipped over 60 reviewed pull requests.

FlowForge — serverless automation service · Next.js, TypeScript, Upstash QStash

Separates frontend events from asynchronous background jobs with durable delivery and retry.

Turbin3 · Solana — Q4 '25 Builder, Q1 '26 Accelerate · Rust, Anchor

Selected for both cohorts. Built NFT marketplaces, staking protocols, vaults, escrow systems and AMMs using PDAs, CPIs and SPL tokens.

Homeworker — local-first document pipeline · Python

PDF and image input to reviewable A4 notes in licensed handwriting typefaces. Public repo, v0.2.0.

MacFolio — browser-based macOS simulation · React, TypeScript, Zustand, GSAP

Draggable and resizable windows, Finder, Safari, Gallery and Terminal with window-manager state.

SKILLS

Languages Rust (working knowledge of compiler internals), TypeScript, Python, Go, Solidity

Engineering Next.js, React, Node.js, Anchor, Prisma, PostgreSQL, Redis, Docker, Linux, GitHub Actions

Automation n8n, monday.com, Upstash QStash, REST, webhooks, OAuth, Playwright, async job design

Systems Microsoft 365 / Exchange, ThreatDown Nebula, FortiClient / FortiToken, RDP, Windows VMs

EDUCATION

Sister Nivedita University — B.Tech, 2023–2027 · Kolkata

National English School — ISC Science (PCM), 85%, 2023